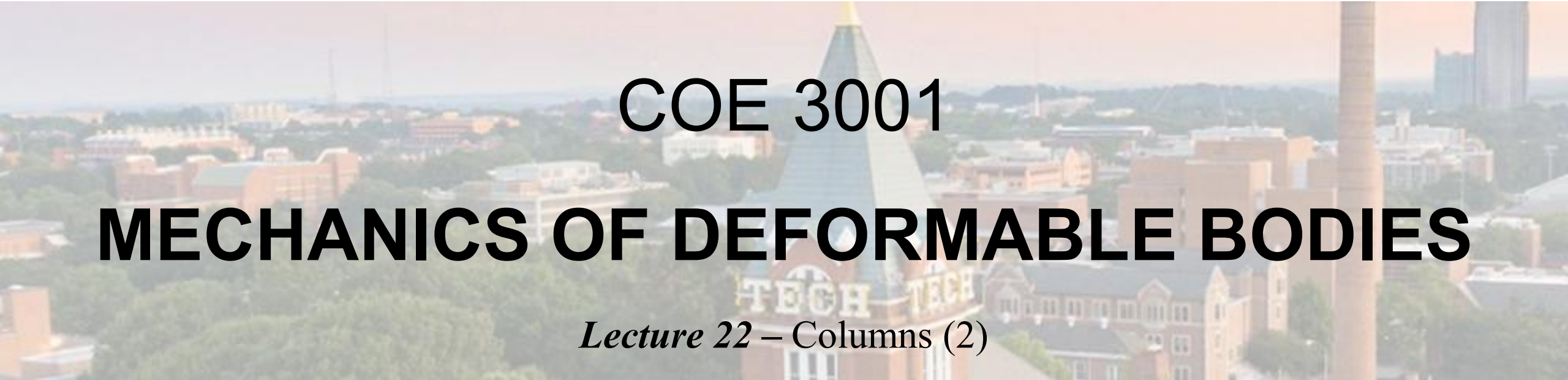




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Engineering

A wide-angle, slightly hazy photograph of the Georgia Institute of Technology campus. The image shows various brick buildings, green trees, and a prominent tall brick smokestack on the right side. The text is overlaid on this background.

COE 3001

MECHANICS OF DEFORMABLE BODIES

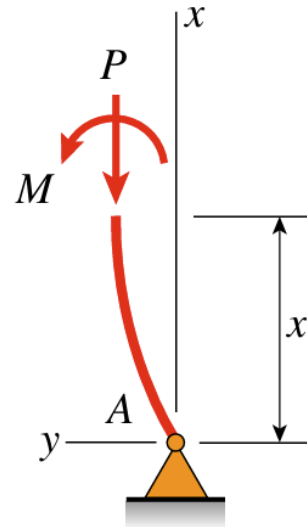
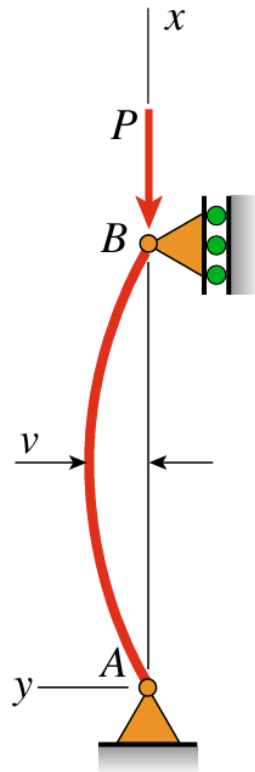
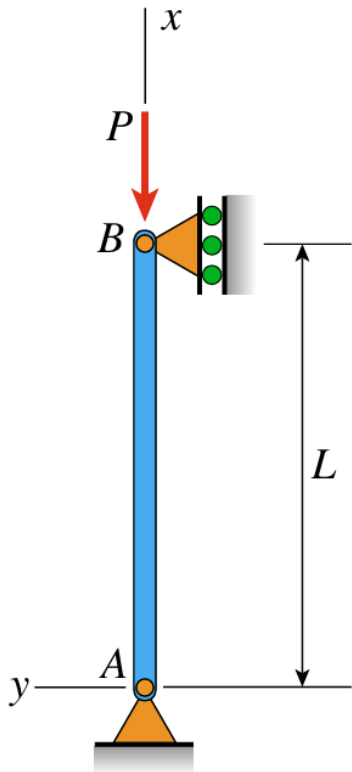
Lecture 22 – Columns (2)

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Differential equation for column buckling



For an ideal pin-ended column

$$EIv'' = M$$

Moment equilibrium

$$M = -Pv$$

Gives

$$EIv'' + Pv = 0$$

Solution of the differential equation

$$EIv'' + Pv = 0$$

Let $k^2 = P/EI$, then

$$v'' + k^2v = 0$$

Seek $v = e^{rx}$ for the solution

$$r^2e^{rx} + k^2e^{rx} = 0 \Rightarrow r = \pm ik$$

The general solution (complex case)

$$v(x) = C_1e^{ikx} + C_2e^{-ikx}$$

In real case

$$v(x) = A \sin kx + B \cos kx$$

Boundary conditions

$$v(0) = 0 \Rightarrow B = 0$$

$$v(L) = 0$$

$$\Rightarrow A = 0$$

$$\Rightarrow \sin kL = 0$$

$$\Rightarrow kL = 0, \pi, 2\pi, \dots$$

Not
curved

Euler's formula for pin-ended beams

$$\sqrt{\frac{P}{EI}} L = 0, \pi, 2\pi, \dots$$

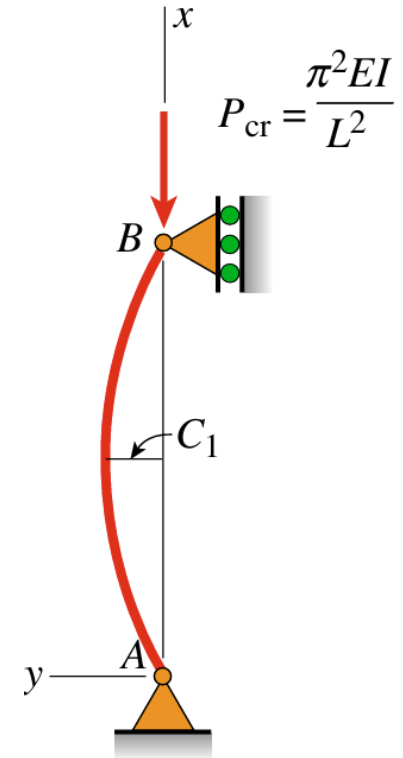
When $p \neq 0$, the minimal solution $kL = \pi$ corresponds to the minimum critical force, which is the engineering critical force

$$\sqrt{\frac{P_{\text{cr}}}{EI}} L = \pi \Rightarrow P_{\text{cr}} = \frac{\pi^2 EI}{L^2}$$

$$v(x) = A \sin \frac{\pi x}{L}$$

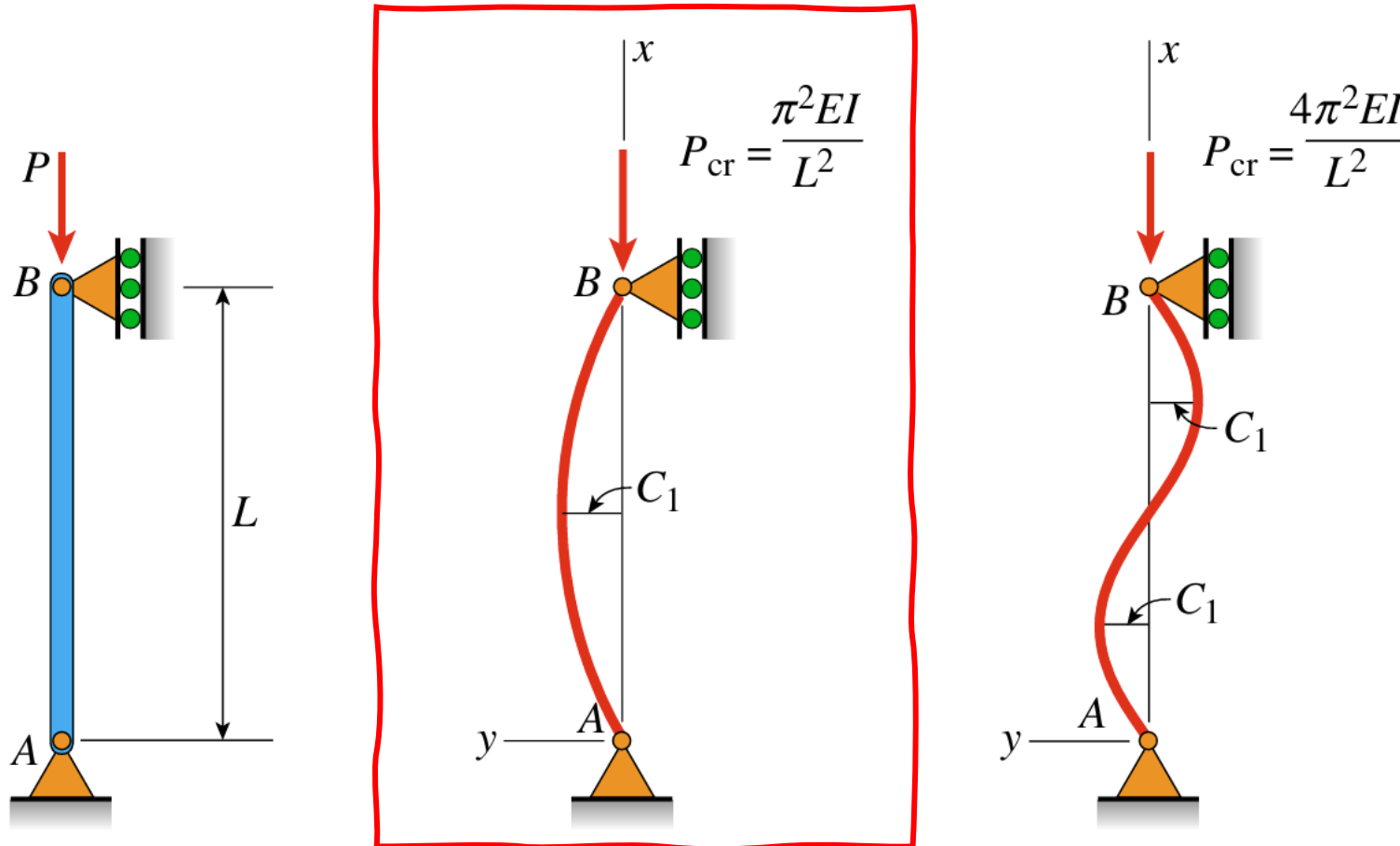
Euler's formula for
pin-ended beams

Critical deflection
curve



Euler buckling

Buckling of a pinned-end column in the first mode is called **the fundamental case of column buckling**.



Critical stress

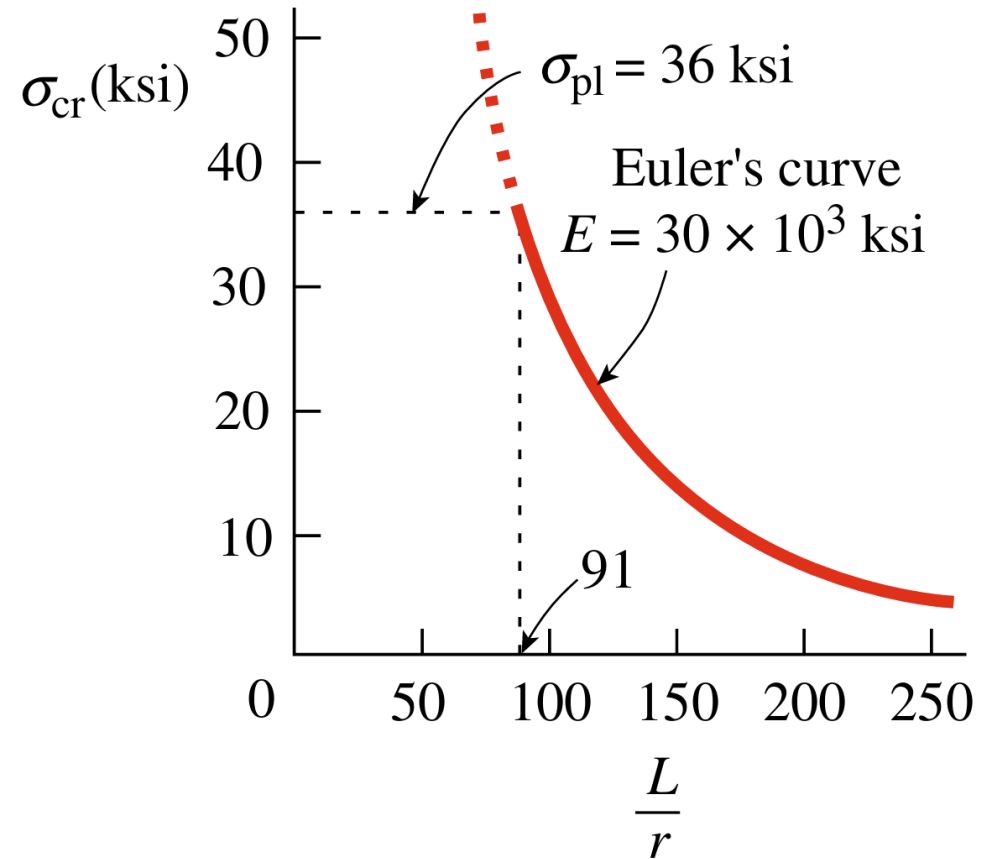
$$\sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 EI}{AL^2}$$

This equation can be written in a more useful form by introducing the notation

$$r = \sqrt{\frac{I}{A}} \Rightarrow \sigma_{cr} = \frac{\pi^2 E}{(L/r)^2}$$

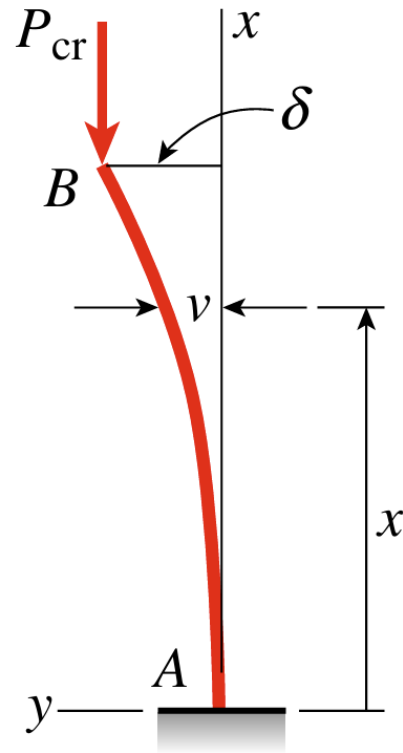
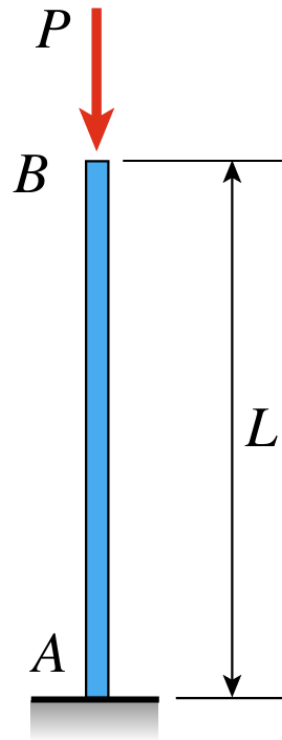
Slenderness ratio

$$\lambda = L/r$$



Extension of Euler's formula

1. Fixed-free beams



$$M = P(\delta - v)$$

$$EIv'' = M = P(\delta - v)$$

Using $k^2 = \frac{P}{EI}$

We have

$$v'' + k^2v = k^2\delta$$

Homogeneous solution:

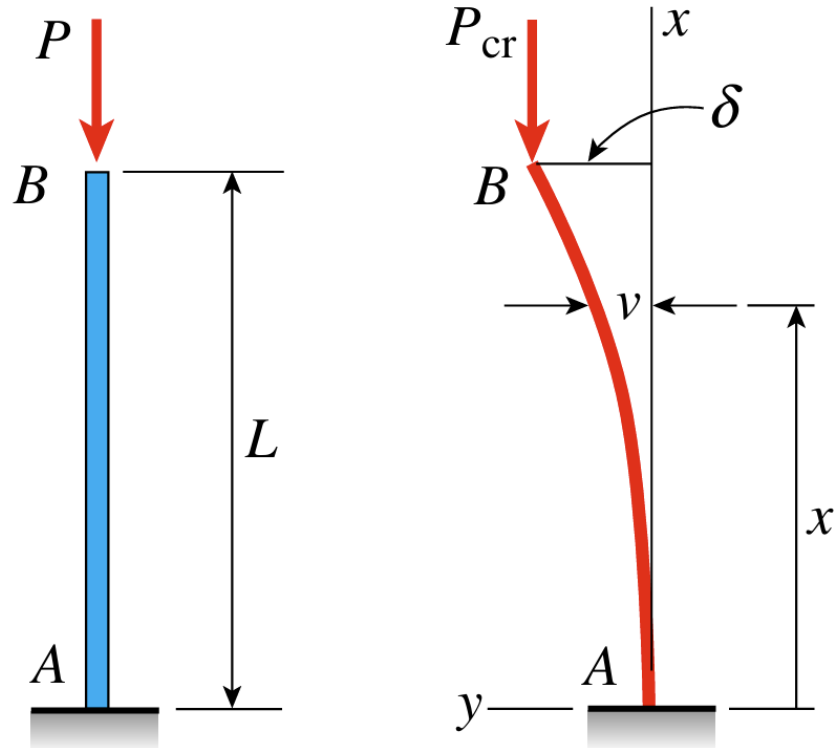
$$v_H(x) = C_1 \sin kx + C_2 \cos kx$$

Particular solution:

$$v_P(x) = \delta$$

Extension of Euler's formula

1. Fixed-free beams



$$v(x) = C_1 \sin kx + C_2 \cos kx + \delta$$

Boundary conditions

$$v(0) = 0 \Rightarrow C_2 = -\delta$$

$$v'(0) = 0 \Rightarrow C_1 = 0$$

So

$$v(x) = \delta(1 - \cos kx)$$

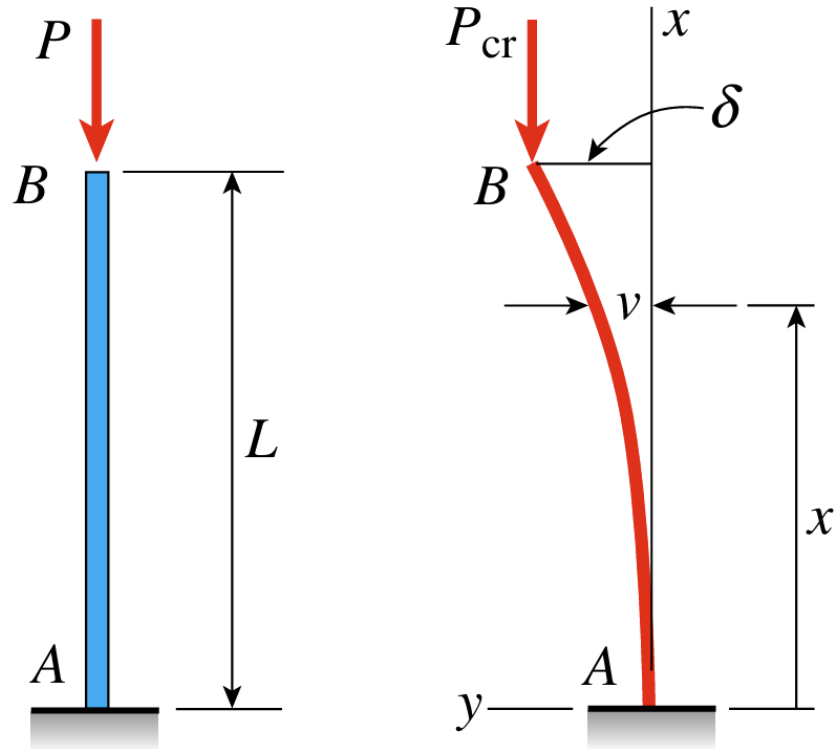
$$v(L) = \delta$$

$$\Rightarrow \cos kL = 0$$

$$\Rightarrow kL = \frac{n\pi}{2}, n = 0, 1, 2, \dots$$

Extension of Euler's formula

1. Fixed-free beams

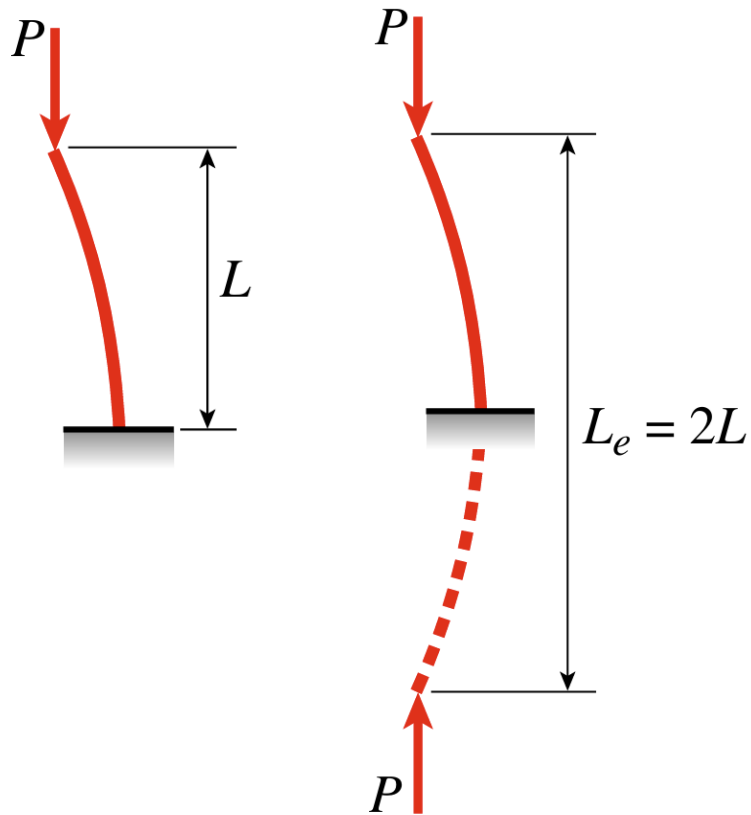


The first mode:

$$P_{cr} = \frac{\pi^2 EI}{4L^2}$$
$$v = \delta \left(1 - \cos \frac{\pi x}{2L} \right)$$

Extension of Euler's formula

1. Fixed-free beams



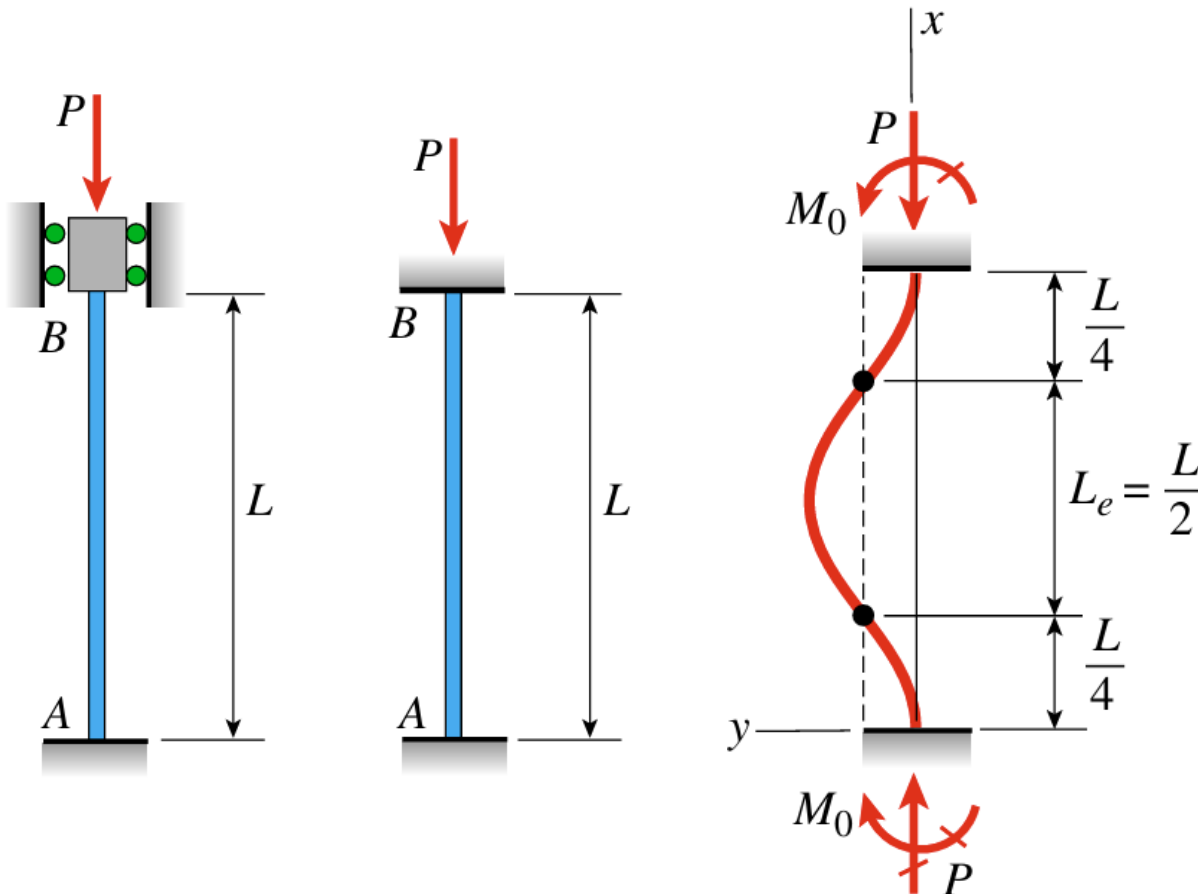
- The **effective length** L_e for any column is the length of the equivalent pinned-end column, that is, it is the length of a pinned-end column having a deflection curve that exactly matches all or part of the deflection curve of the original column.
- The effective length is often expressed in terms of an **effective-length factor** K

$$L_e = KL$$

$$P_{cr} = \frac{\pi^2 EI}{L_e^2} = \frac{\pi^2 EI}{(KL)^2}$$

Extension of Euler's formula

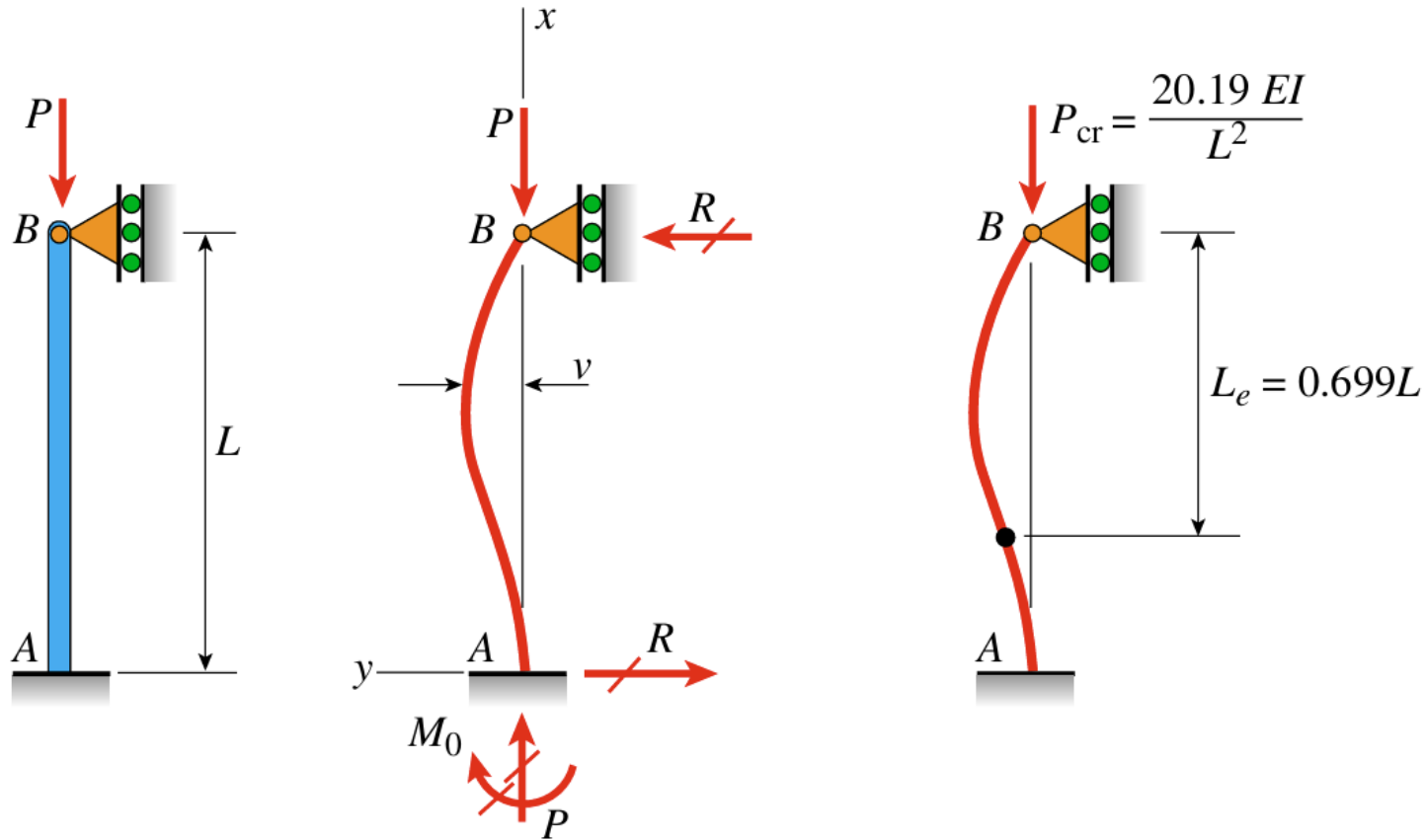
2. Fixed-fixed beams



$$P_{\text{cr}} = \frac{4\pi^2 EI}{L^2}$$
$$K = 0.5$$

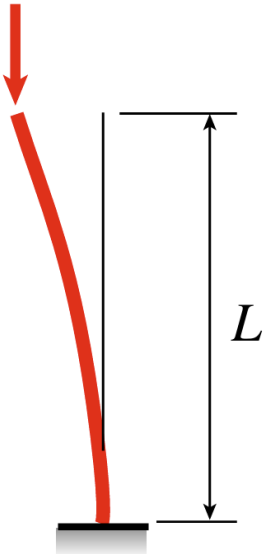
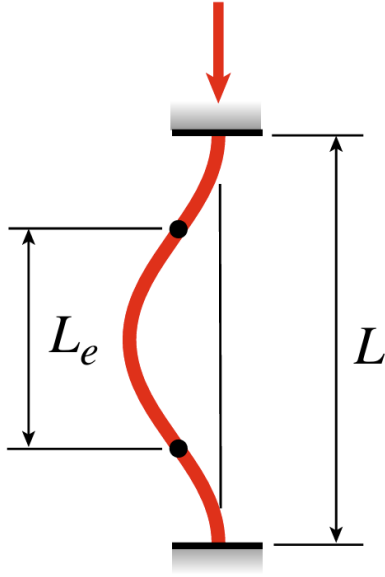
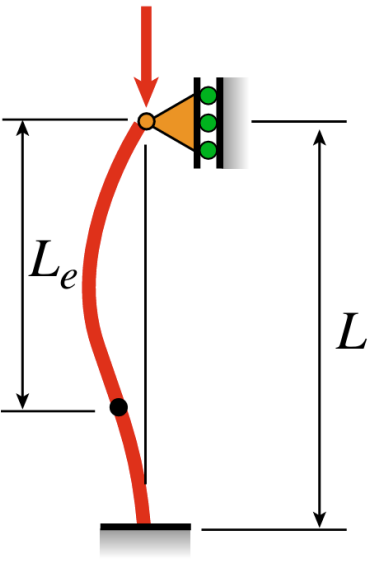
Extension of Euler's formula

3. Fixed-pinned beams



$$P_{cr} = \frac{2.046\pi^2 EI}{L^2}$$

$$K \approx 0.7$$

(a) Pinned-pinned column	(b) Fixed-free column	(c) Fixed-fixed column	(d) Fixed-pinned column
$P_{cr} = \frac{\pi^2 EI}{L^2}$	$P_{cr} = \frac{\pi^2 EI}{4L^2}$	$P_{cr} = \frac{4\pi^2 EI}{L^2}$	$P_{cr} = \frac{2.046 \pi^2 EI}{L^2}$
			
$L_e = L$	$L_e = 2L$	$L_e = 0.5L$	$L_e = 0.699L$
$K = 1$	$K = 2$	$K = 0.5$	$K = 0.699$

Euler's formula

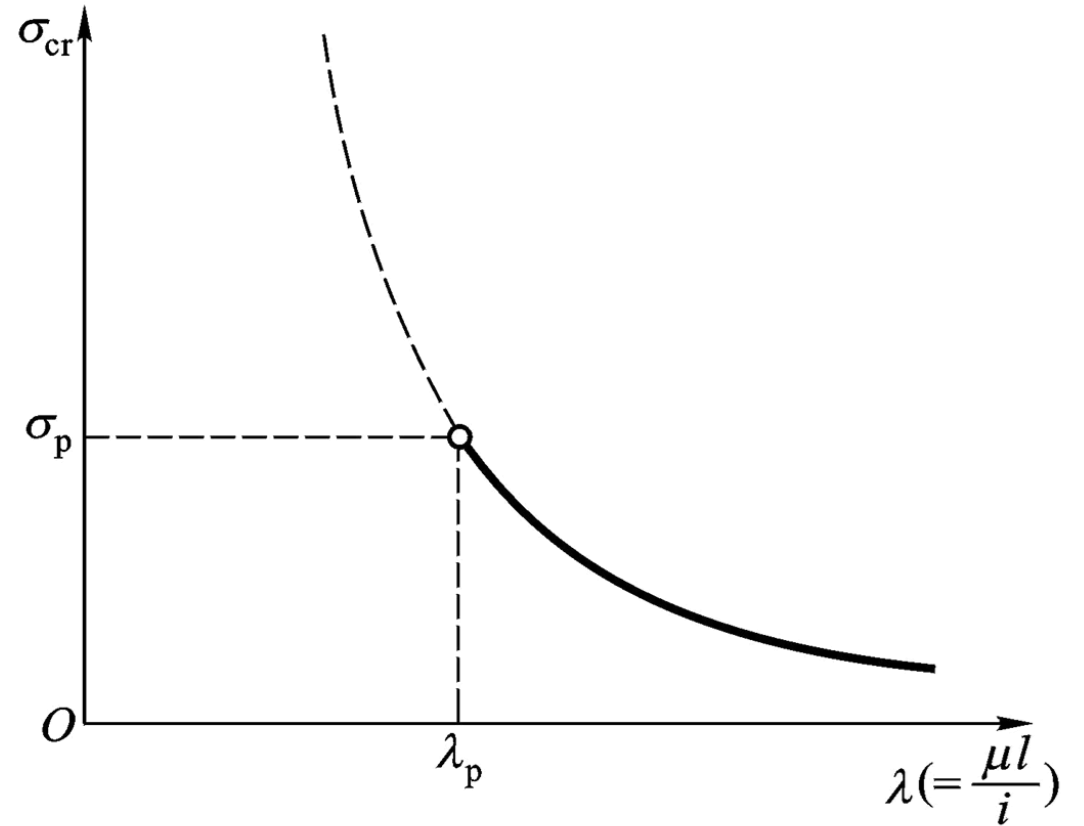
$$P_{\text{cr}} = \frac{\pi^2 EI}{(KL)^2}$$

- K : effective-length factor
- KL : effective length

Applicable range

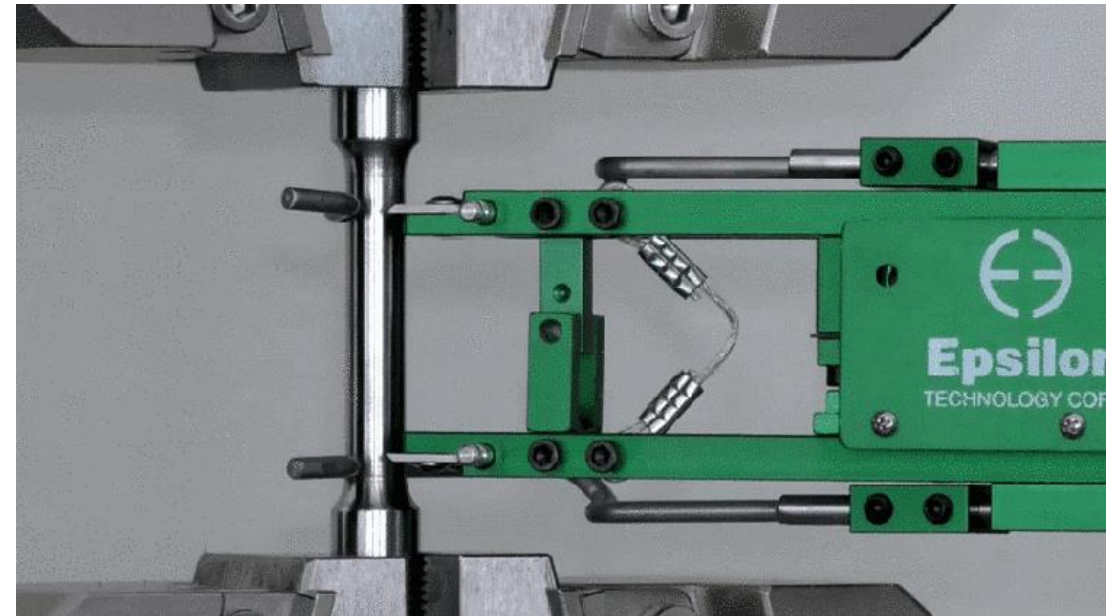
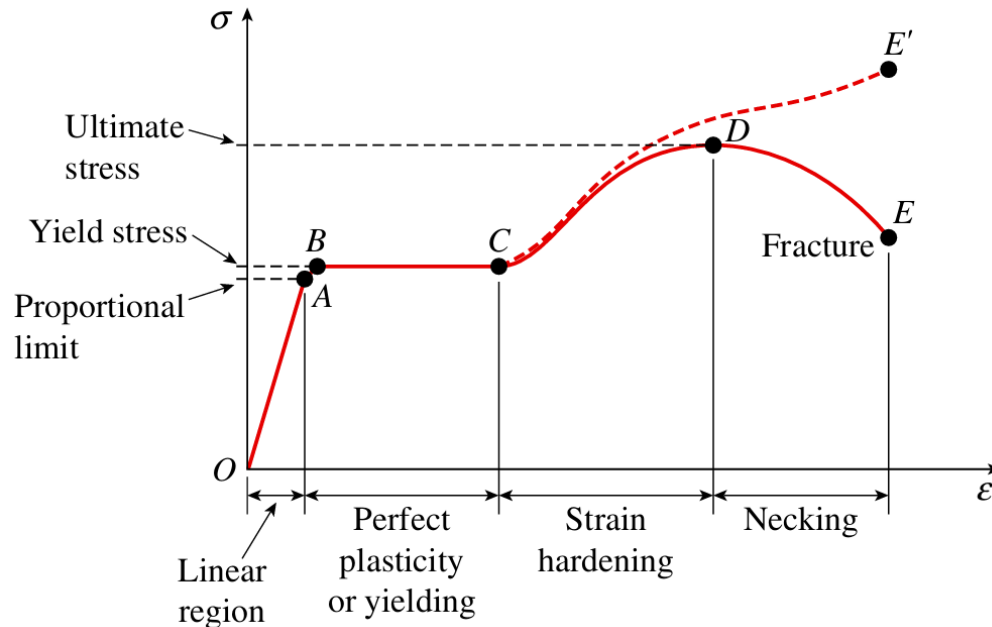
$$\sigma_{\text{cr}} = \frac{\pi^2 E}{\lambda^2} = \frac{\pi^2 E}{(KL/r)^2} \leq \sigma_p \Rightarrow \lambda \geq \lambda_p = \sqrt{\frac{\pi^2 E}{\sigma_p}}$$

Slenderness ratio



Stresses and Strains for Structural Steel

Structural steel, also known as *mild steel* or *low-carbon steel*, is one of the most widely used **ductile** metals found in buildings, bridges, cranes, ships, towers, vehicles, and many other types of construction.



Safety factor for stability

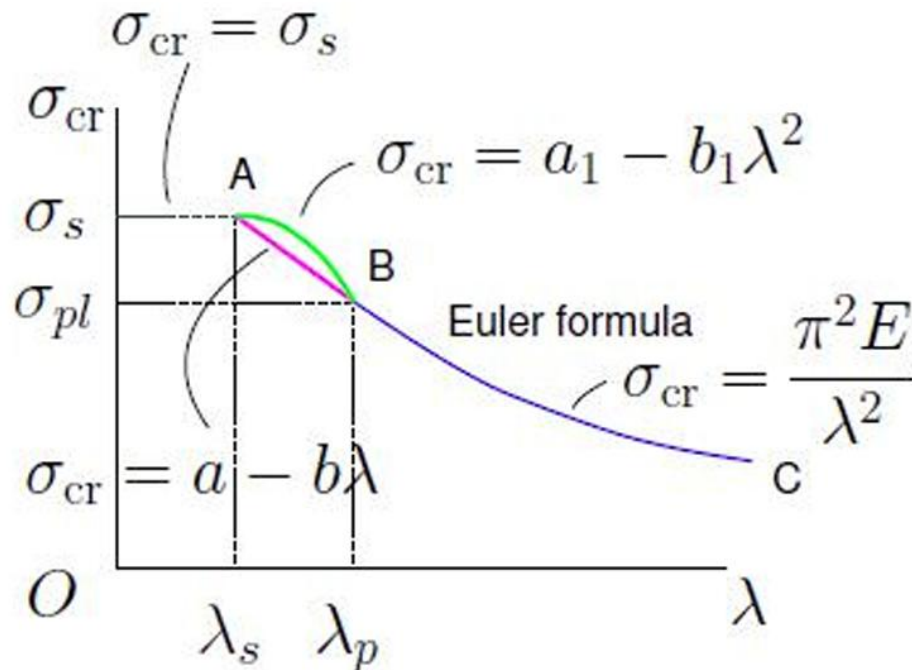
Safety factor stability n_{st}

$$\frac{F}{A} \leq \frac{\sigma_{cr}}{n_{st}} = [\sigma]_{st}$$

$[\sigma]_{st}$ is called **stability allowable stress**

Limitations of Euler buckling

Derivation of Euler buckling requires linear behavior, $\lambda \geq \lambda_p$. For $\lambda < \lambda_p$, Euler buckling is invalid.



Proportional limit

Strength

Empirical formulas for $\sigma_{pl} < \sigma_{cr} < \sigma_s$

- Linear relation

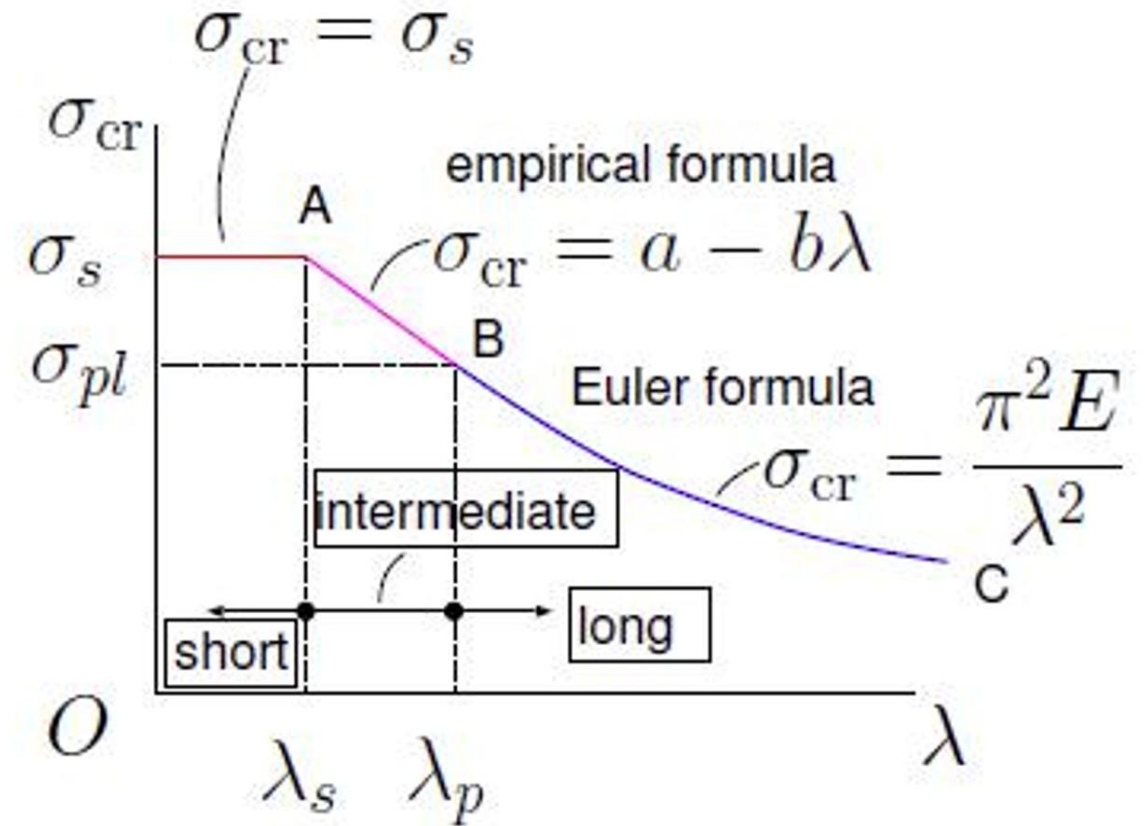
$$\sigma_{cr} = a - b\lambda$$

- Parabolic relation

$$\sigma_{cr} = a_1 - b_1\lambda^2$$

Inelastic buckling

- Long columns ($\lambda > \lambda_p$): Euler formula
- Intermediate columns ($\lambda_s < \lambda < \lambda_p$): empirical formula
- Short columns ($\lambda < \lambda_s$): too short to buckle non-stability issue Strength theory



$$\lambda_p = \sqrt{\frac{\pi^2 E}{\sigma_{pl}}}, \quad \lambda_s = \frac{a - \sigma_s}{b}$$